

Dominic Saksa

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Computer Engineering student pursuing a career in software engineering and embedded systems with experience in embedded programming, independent PCB design, and team-based application development.

EDUCATION

Michigan State University - East Lansing, MI
Bachelor of Science, Computer Engineering

August 2023 - expected May 2027

Yonsei University - Seoul, South Korea
Summer Semester Abroad

Summer 2024

TECHNICAL SKILLS

Programming:

C#, C++, CSS, HTML, JavaScript, Python, Rust, SPICE, Verilog

3D Modeling, CAD, and Game Dev:

AutoCAD, Blender, CATIA V5, KiCad, Unity

WORK EXPERIENCE

Xiomas | Embedded Systems Intern

July 2026 - August 2026

- Designed and implemented C++ drivers for a thermal imaging camera, integrating UVC video streaming and serial communication for camera control and data acquisition.
- Developed and debugged low-level communication interfaces between the camera and host system, translating device protocols into software-accessible controls and telemetry.
- Integrated and tested the camera subsystem within the broader imaging pipeline, troubleshooting hardware/software communication issues and validating reliable operation.

Bloxxide | Context Engineering Intern

May 2026 - August 2026

- Evaluated and benchmarked AI agent stacks (Kimi K2P6/K2P7, OpenClaw, OpenCode, Hermes) across speed, accuracy, oversight, and task completion to select the optimal internal development stack.
- Deployed Hermes Agent on a remote Linux server via Tailscale mesh VPN with Fireworks AI inference, and built a four-stage Rust CLI pipeline generating daily DFM/DFA content from IPC-2221/2222 sources with JSONL audit traceability via scheduled cron jobs.

BI Worldwide | Junior Developer

June 2025 - May 2026

- Developed responsive HTML email campaigns for Metro Detroit automotive dealerships, converting PDF templates into StrongMail-ready packages for employee incentive communications.
- Applied responsive design across major email clients and collaborated with account teams to align with brand guidelines.

PROJECTS

Spartanoid | Object-Oriented Software Design - C++

- Led a team building a multi-level brick-breaker game with real-time physics, collision detection, and scoring.
- Engineered a polymorphic game object architecture with the Visitor design pattern for heterogeneous object interactions and XML-based level loading for dynamic configuration.

ALU Project | Microprocessors and Digital Systems - Verilog and Vivado

- Programmed an 8-bit ALU with four logic operators and multiplexer output selection

ATM State Machine | Computer Logic Fundamentals - Verilog and Vivado

- Designed a finite state machine controlling an ATM with checking, savings, deposit, withdrawal, transfer, and bill pay. Interfaced with simulated hardware using 8-bit balances, transaction commands, and warning indicators on a 100 MHz clock.

Pokémon Parallax Project - Independent Project

- Built a head-tracking system using a PS3 Eye camera, OpenTrack, and Unity to make a 2D screen appear 3D as the player moves their head.
- Implemented UDP network transmission of tracking data and mapped it to 3D coordinate space using Off-Axis projection, simulating perspective through a window.

LEADERSHIP & COLLABORATION

Triangle Fraternity | MSU Chapter - FIRST Robotics Chair

IEEE | MSU Branch - Member

Talk of the Irish - Founder & Leader

- Established and led the Notre Dame Preparatory podcast team to amplify the student voice.